Typed, Calibrated Decisions for an Offline Network-Management Agent:

Serving the System One Idea Next to a 4B Model on an Embedded NPU

Krish Viswanathan

September 2026 — preprint, v0.1

Abstract

An offline network-management assistant runs a 4-billion-parameter instruction-tuned model on an embedded NPU (Qwen3-4B-Instruct-2507, W4A16, Qualcomm QCS9075) and turns an operator's sentence into one tool call against a managed PoE switch; a human confirms every write. The generative path is accurate on canonical phrasing (92–97% end-to-end) and weaker on paraphrases (77.4%; 61.5% on indirect phrasing), but two properties matter more than the headline: its misses are structured, several of them dangerous (cut PoE power versus shut the interface; one access VLAN versus a trunk), and it emits no probability, so the confirm gate cannot tell a confident call from a coin flip. On the compiled NPU serving path there are no logits to read. We borrow the interface of TypeSafe's Jev — typed questions (yes/no, one of K, ordered levels) over a state, answered in one pass with calibrated probabilities — and serve it locally on the CPU next to the generative agent. We compare four realizations on one labeled workload of 519 operator requests (65 seeds, 195 style-controlled paraphrases, 259 rows authored for the study): a next-token readout of the resident 4B over single-token labels via `llama-server`, three open encoder decision models zero-shot (151M–421M), a 150M encoder fine-tuned on the domain, and rules. We measure accuracy per phrasing style, calibration before and after out-of-fold temperature scaling (folds grouped by seed request), order sensitivity under option rotation, selective accuracy at the thresholds Jev recommends, and how many of the generative model's *real* misses a "does this proposal match the request" question flags. The readout is not a better router (84.6% on a 24-way command question vs the generative call's 83.1%), and its raw confidence is unusable (23 of 40 errors above $p = 0.9$; 29–36% of answers flip under option reordering on adversarial rows), but once temperature-scaled and rotation-averaged it gives the gate an AUROC-0.95 check that flags 15 of the agent's 20 real dangerous confusions at $p < 0.5$ while false-flagging 30 of 151 correct proposals, and admits 1 of 156 dangerous synthetic swaps; on the board's CPU it costs 1.3 s per router question and 3 s per gate question with the question prefix cached. The zero-shot open reproductions are at chance on the gate and collapse above about six options. A 150M encoder fine-tuned on $\sim$2,000 rows with the question order shuffled becomes the better router than the readout on identical held-out rows (tool 88.1% vs 84.9%, command 77.8% vs 69.8%) at 126 ms per question on the board's CPU, but no encoder variant learns the gate (AUROC 0.53–0.65): the division of labour is an encoder for the per-turn router and a decoder readout for the per-write gate. We release the code, the dataset and the result receipts.

1 Introduction

Small language models now drive tool-calling agents on embedded accelerators. On a Qualcomm QCS9075 board, a 4B instruction-tuned model compiled to the Hexagon NPU answers an operator in 6–7 s per turn and reaches 92–97% end-to-end accuracy on a 65-case tool-calling benchmark against a live managed switch [14]. That accuracy is measured on canonical phrasing. On a 195-case paraphrase suite the same deployment scores 77.4%, and 61.5% when the operator states a symptom instead of a command.

The product is protected by a confirm gate: no configuration write reaches the switch until a human approves the rendered command. That gate is only as good as what it shows. Today it shows the proposed call and the model's free-text reason, both written by the model that may have misread the request. It cannot show a probability, because a generated tool call is a one-hot event, and it cannot read one off the serving stack, because the compiled NPU path exposes no logits (§6).

Two observations about the misses motivate this study. First, they are structured: of the 44 paraphrase misses, 20 are confusions between semantically adjacent catalog commands (12 access-VLAN vs trunk, 4 VLAN config vs membership, 2 PoE-off vs port shutdown, 1 speed vs shutdown, 1 PoE mode vs limit), 13 are "run diagnostics instead of a status read" (usually defensible), 10 are format slips (a bare command name in the tool slot, recoverable by a deterministic normalization) and 1 is other. Second, the dangerous subset is small in number and large in consequence: shutting an interface when the operator asked to cut PoE power takes a camera off the network until someone reverts it.

TypeSafe's Jev [11, 13] reframes decisions like these as *typed questions* over a *state*: a NOUL (yes/no) returns one probability, a CHOICE returns a distribution over supplied options, a SCORE returns a distribution over ordered levels and its expected value; all questions are answered in one non-autoregressive pass, nothing is generated, and the model is trained ("RLCD") so that the probabilities are calibrated. Whether or not one uses that particular model, the *interface* is the right object for a confirm gate: a distribution the application thresholds, with a confidence axis that a text tool call lacks. Jev is cloud-only; an air-gapped board cannot call it. Within days of its release several open reproductions appeared [5, 8, 6, 1, 3]; they are small encoders (151M–421M) or LoRA-adapted decoders, and they run on a CPU.

This paper asks a concrete question: *how much calibrated, typed decision-making can be served locally, next to the generative agent, for this workload, and what does it buy the confirm gate?* We make four contributions.

1. **A labeled decision dataset for a real edge agent.** 519 operator requests to a managed PoE switch (65 seeds, 195 style-controlled paraphrases, and 259 rows authored for this study: adversarial confusables, other intents, out-of-scope), each with the correct tool call and typed labels (is a write, touches PoE power, ambiguity 0–3, acceptable alternatives, blast radius), plus the generative model's actual calls on the same requests.

2. **Four local realizations of the same interface** behind one library (`jevlet`): a next-token probability readout of the resident 4B over single-token labels with prefix caching and option-rotation averaging; three open encoder decision models zero-shot; a small encoder fine-tuned on the domain with a proper-scoring-rule loss; and rules. All return the same object (a distribution plus its pre-softmax scores), so calibration is measured the same way for each.

3. **Measurements that a confirm gate can act on:** accuracy per phrasing style, raw and out-of-fold temperature-scaled calibration with folds grouped by seed request, order-flip rate under option rotation, selective accuracy at Jev's act/confirm/human thresholds, the fraction of the generative model's *real* misses that a "does this proposal match the request" question flags, and CPU latency at one and four threads.

4. **Negative and serving-path results** that bound the design space on this class of hardware: a text-label choice from the same 4B is *less* accurate than its own tool call; the compiled NPU runtime returns no logprobs on any version; and raw readout probabilities are unusable as confidence until scaled (27 of 32 errors above $p = 0.9$ on a 15-way question).

2 Background

System One models and Jev. Jev [11] is trained with "Reinforcement Learning for Calibrated Decisions" rather than RLHF, evaluates every question against one state in parallel in 70–500 ms, and returns per question a probability distribution and a concentration statistic, *confidence* $= (K \max_i p_i - 1)/(K - 1)$ [13]. Its documentation recommends acting above confidence 0.9, confirming between 0.5 and 0.9 and escalating below, with the threshold raised for destructive actions, and documents "jagged edges": literal reading, weak numeric and date reasoning, accuracy loss as irrelevant state grows, and no resistance to instructions embedded in the state [12].

Open reproductions. openJev-verdict-2.0 [5] is a 151M ModernBERT-base GLiClass cross-encoder with an explicit abstention slot, a CE+Brier loss and temperature scaling, shipped as ONNX. Laya [8] is a 421M ModernBERT-large with option-marker scoring trained by a GRPO-style policy gradient against proper scoring rules. open-jev-deberta-v3-large [6] scores span means of question and option tokens with a small head over a DeBERTa-v3-large encoder, trained on public gold labels only; its authors report an in-domain/out-of-domain gap of 0.85 vs 0.69 accuracy. Open-Jev-2B/9B [1] attach a LoRA adapter and a scalar head to Qwen3.5 decoders. OpenJev [3] is a readout of candidate log-likelihoods from any instruction-tuned LM with KV-shared prefixes and temperature scaling, wire-compatible with the Jev SDK. We use the first three zero-shot and the readout idea in our own implementation.

Calibration and selective prediction. Expected calibration error, Brier score and negative log-likelihood measure whether stated probabilities match observed frequencies; temperature scaling [4] is the standard post-hoc fix; selective classification reports error as a function of coverage at a confidence threshold [2]. Instruction-tuned language models are known to be miscalibrated after RLHF and sensitive to the order and labels of multiple-choice options [16, 10]; option permutation averaging is a training-free mitigation.

Tool calling on edge devices. Tool-calling benchmarks are cloud/GPU-side [9]; on-device work reports tokens per second, not agent accuracy. The deployment studied here was characterized in [14]: the serving path (chat template fidelity, tool-call parsing) moved the same weights between 0% and 100% tool-calling accuracy, and a compiled context window prices capacity in per-token time.

3 The workload and the dataset

The agent. The application is a network device manager with a built-in assistant. On the board the assistant is a single stateless turn: a 900-token system prompt, the JSON schemas of a 4-tool profile (`query_device_info`, `configure_device_setting`, `diagnose_port_issues`, `search_documentation`; the other eight tools do not fit the 4K compiled context), the request, one generation, a bounded tool loop, and, for any write, a confirmation dialog rendering the exact CLI. The two wrapper tools carry a `command_name` enum over a 23-command catalog.

Rows. We start from the application's evaluation set — 65 canonical requests in four difficulty levels and 195 paraphrases in three styles (A casual, B verbose, C indirect / symptom-first) that preserve the seed's tool, command and parameters — and add rows written for this study: 81 adversarial confusables (style X) that sit between adjacent commands ("kill power on 3", "take port 3 down", "not a power issue — just kill the link on 6", "stick port 9 on vlan 100", "knock port 8 down to 10"), 54 symptom-first diagnostics requests, 76 documentation, out-of-scope and fleet-wide messages, 48 requests for tools outside the production profile, and 983 label-preserving training paraphrases (A/B/C plus a terse/typo style T) used only for fine-tuning.

Every authored row carries `is_write`, `touches_poe_power`, `ambiguity` (0–3; 35 rows at 2, 1 at 3), `alt_ok` and `blast` labels; 46 rows stress PoE-off vs shutdown, 20 access vs trunk, 13 PoE mode vs limit, 10 speed vs shutdown, 9 VLAN config vs membership, 28 status vs diagnose. Rows were generated by one LLM author and checked by an independent LLM verifier for label correctness, format, realism and duplicates (dropping near-duplicates and rows that contradicted the canonical convention); the change log is published with the data, and human review is pending. All evaluation splits are grouped by seed request.

Question batteries. The *router* battery is asked of the request alone: `tool` (CHOICE over 8, including *none*), `command` (CHOICE over the 23 commands plus *none*; the confusable pairs are named in the instructions), `is_write` (NOUL), `touches_poe_power` (NOUL). The *gate* battery is asked of the request together with a proposed call and its one-line description: `matches_request` (NOUL). Gate labels come from three sources: the generative model's *real* calls on the 195 paraphrases (label = its graded end-to-end correctness; 151 positive, 44 negative), the gold call (positive), and synthetic negatives that swap the operation for its confusable neighbour or change the port: 939 pairs (195 real, 260 gold, 260 swapped operation, 224 wrong port), of which 156 are "dangerous" swaps (PoE-off↔shutdown, access↔trunk, speed↔shutdown, PoE mode↔limit). Among the 195 real calls, 34 are semantic misses (20 command confusions, 13 diagnose-instead, 1 parameter) and 10 are format slips whose command and parameters match the request; the latter count as matches for the gate.

4 Four local realizations of one interface

Interface. A request is a state and a map of typed questions; an answer is, per question, a probability vector aligned with the options, the pre-softmax scores it came from (log-probabilities or logits), and the concentration confidence. The application thresholds the probabilities. Every backend below returns exactly this, so one calibration and analysis pipeline serves all of them.

Next-token readout (`llamacpp`). The question — instructions, the option legend with single-token labels (A..X for choices, 0..9 for score levels, Yes/No for NOUL) and the answer format — is the system message; the state is the user message; we request one token with `logprobs` and read the pre-sampling distribution at the first generated position, summing case and whitespace variants and renormalizing over the label set. Mass on non-label tokens is reported as *label mass*; here it was 1.000, i.e. the model always emitted a label. Because the system message is identical across states, consecutive requests reuse the server's prefix cache and cost only the state's tokens. Order sensitivity is reduced by evaluating R cyclic rotations of the option list and averaging in option space. This needs a llama.cpp server (CPU, CUDA or Hexagon); it cannot target the compiled NPU runtime.

Open encoder decision models (zero-shot). openJev-verdict-2.0 (ONNX, CPU), Laya (English and the typed-decisions checkpoint) and open-jev-deberta-v3-large, each called through its own SDK with the same instructions and option descriptions. One forward pass answers all router questions for Laya and Verdict; the DeBERTa checkpoint's 512-token context forces one question per pass for the 24-option question.

Fine-tuned encoder. ModernBERT-base and DeBERTa-v3-base trained with the typed-decisions recipe [6]: the encoder reads `[STATE] state [Q] instructions [OPT] option ...`, a small head scores the span means of question and option tokens, softmax within each question, loss = cross-entropy + Brier, with gold-preserving augmentation (option shuffling, distractor dropping, negation with flipped gold). Trained on the training groups only; evaluated on held-out groups; temperature fitted on a validation split.

question	acc.	canon / A / B / C	ECE raw→T	wrong@conf>0.9 raw→T	T
tool (8-way)	96.5	100 / 95.4 / 100 / 90.8	0.033→0.013	4→6 of 9	3.3
command (24-way)	84.6	87.7 / 75.4 / 89.2 / 86.2	0.133→0.070	23→12 of 40	3.4
is_write	98.1	100 / 98.5 / 100 / 93.8	0.019→0.008	4→1 of 5	4.0
touches_poe_power	98.8	100 / 96.9 / 98.5 / 100	0.014→0.010	3→3 of 3	4.1

Table 1: Readout (Qwen3-4B-Instruct-2507 Q4_0, `llama-server`, 3 rotations) on the 260 canonical + paraphrase requests. Confidence is Jev's concentration statistic.

Rules. Regexes and lookup tables (does the request name a port? is it multi-device? the deterministic map from a proposed operation to its semantic class). One-hot output; the floor every deployment gets, and the fallback on a model timeout.

Where the layer sits. Before the generative turn, the router battery selects which tool schemas to load into the 4K context and whether to add a one-line hint to state the reading explicitly; after the generative turn, the gate battery annotates the confirm card with a probability and a plain sentence ("you asked to cut PoE power; this proposal shuts the whole port down"). The layer never executes. Jev's "act automatically above 0.9" is deliberately not applied to device writes.

5 Experiments

All model runs used greedy decoding or a single forward pass; hosts were an aarch64 Grace/GB10 machine (llama.cpp commit cb295bf, CPU build; torch 2.14, transformers 5.17) and the QCS9075 board (llama.cpp Snapdragon package, CPU-only with the Hexagon device count set to zero; Python 3.13 with CPU torch 2.10 in an isolated environment). Calibration is scalar temperature scaling fitted by NLL on four of five folds grouped by seed request and applied to the fifth; every number below is out-of-fold. Result receipts for each table are in the repository.

5.1 E1 — the readout of the resident 4B

Table 1 gives the router battery on the 260 canonical + paraphrase requests (3 rotations averaged). Three observations. (i) The readout is not a better router: its 24-way command accuracy (84.6%) is the generative call's semantic command accuracy on paraphrases (83.1%), and it inherits the same confusions (`enable_port`→`disable_port`, speed→duplex). (ii) Raw confidence is unusable: 23 of 40 command errors sit above Jev confidence 0.9; a temperature $T \approx 3.4$ halves ECE (0.133→0.070) and the confident errors (23→12) but cannot remove them. (iii) Order sensitivity: a single-ordering pass on a 15-way variant flipped 15.8% of predictions when the order was reversed; on the 259 authored rows the flip rate across three rotations is 36% (tool) and 29% (command), and rotation averaging adds 4 points of tool accuracy over a single ordering.

On the 259 authored rows the readout drops to tool 68.3%, command 60.2% (64.1% allowing the listed alternatives; 84.3% / 67.4% on the 81 adversarial confusables), `is_write` 85.7%, `touches_poe_power` 95.8%; ECE on the command question falls from 0.261 to 0.041 with $T \approx 4.9$, at which point no answer clears 0.9. The most frequent miss is mapping out-of-scope messages onto `show_cable_diagnostics` (25 rows); the dangerous pair PoE-off→`disable_port` accounts for 8.

The gate. On the 939 pairs the readout reaches AUROC 0.952 (ECE 0.108→0.030, $T \approx 6.4$). Synthetic negatives are nearly airtight: swapped operations are accepted 7.3% of the time, wrong ports never, and 1 of

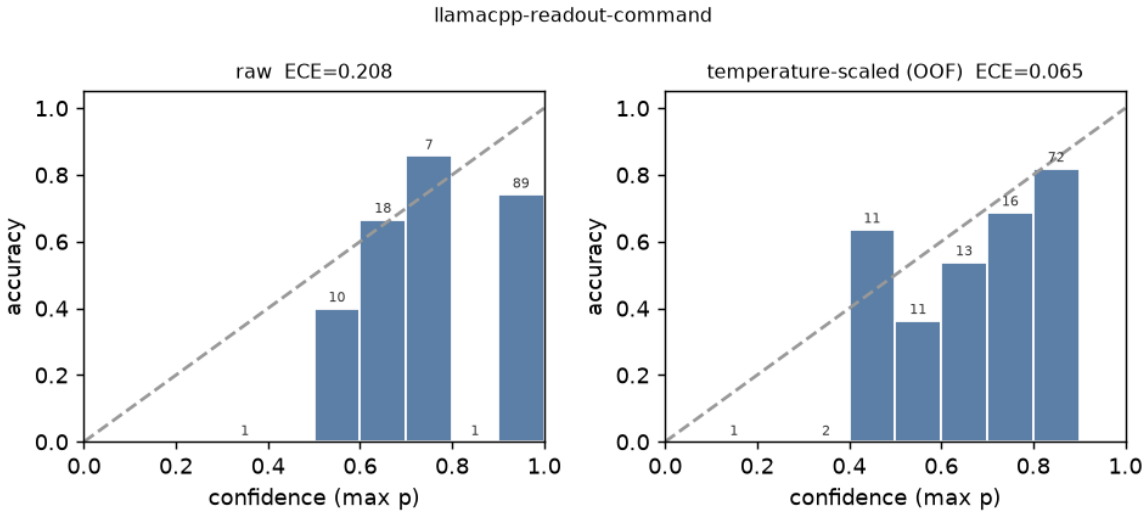


Figure 1: Reliability diagrams of the readout's 24-way command answers on the held-out rows (bin counts above the bars). Left: raw next-token confidence sits at 0.9–1.0 with 74% accuracy (ECE 0.208). Right: one temperature per fold, fitted on other seed requests (T≈4.3), puts the bars on the diagonal (ECE 0.065); no answer clears 0.9 any more.

backend	AUROC	ECE raw→T	confusions flagged	correct false-flagged	dang. swaps accepted
readout Qwen3-4B	0.952	0.108→0.030	15 / 20	30 / 151	1 / 156
readout Qwen3.5-0.8B (no thinking)	0.879	0.119→0.047	13 / 20	36 / 151	25 / 156
Laya 421M (0-shot)	0.757	0.178→0.022	1 / 20	1 / 151	101 / 156
ft. ModernBERT-base, round 1 (held-out)	0.647	0.107→0.062	2 / 2	37 / 40	0 / 38
ft. ModernBERT-base, gate-only (held-out)	0.556	0.050→0.087	1 / 2	14 / 40	8 / 38
open-jev-deberta-v3-large (0-shot)	0.663	0.413→0.070	0 / 20	0 / 151	155 / 156
openJev-verdict-2.0 (0-shot)	0.566	0.295→0.067	0 / 20	0 / 151	156 / 156
NLI cross-encoder (0-shot)	0.547	0.401→0.069	0 / 20	10 / 151	144 / 156

Table 2: The gate question, "does the proposed action do what the operator asked?", at the 0.5 threshold after out-of-fold temperature scaling. Real pairs are the generative model's own calls on the 195 paraphrases.

156 dangerous swaps is accepted at $p > 0.5$. On the generative model's real calls (Table 2) it flags 15 of the 20 command confusions and 7 of the 13 diagnose-instead calls at $p < 0.5$, false-flags 2 of the 10 format slips, and false-flags 30 of the 151 correct calls (76 at $p < 0.9$); AUROC on the real pairs alone is 0.714. Prompt wording moves the raw operating point more than calibration does: an earlier phrasing of the same question put mean $p(\text{yes})$ on correct proposals at 0.21 (AUROC 0.91) against 0.82 for the final phrasing (AUROC 0.95).

5.2 E2 — open encoder decision models, zero-shot

None transfers to the 23-command catalog. openJev-verdict-2.0 answers the 8-way tool question at 72.7% (58.5% with shortened descriptions) but returns the same label for every request on the 24-way command question; probing with 6, 12, 16, 20 and 24 options shows the collapse begins at about six, consistent with its 5-candidate evaluation and with Laya's reported Banking77 weakness. open-jev-deberta-v3-large answers the command question at 70.8% (bare option names, one question per pass because of its 512-token context) and `is_write` at 88.8%, but the tool question at 23.8%. The NLI cross-encoder reaches 53.1% / 86.5%. All three are at chance on the gate (Table 2) and, being honest about it, abstain or spread their mass: after scaling

none accepts a dangerous swap above 0.9. Laya's typed-decisions checkpoint is the strongest zero-shot encoder on all 519 rows (tool 61.7%, command 51.4%, `is_write` 81.1%, `touches_poe_power` 86.3%) and its English checkpoint the strongest zero-shot gate (AUROC 0.757, real pairs 0.710), yet it flags 1 of the 20 real confusions at $p < 0.5$ and accepts 101 of 156 dangerous swaps. On all 519 rows open-jev-deberta scores 31.6 / 45.1 / 75.3 / 39.7 and the NLI baseline 20.6 / 40.8 / 81.9 / 68.2.

A sub-1B decoder. Qwen3.5-0.8B reads out nothing in its default template (the first generated token opens a thinking block: label mass 0); with thinking disabled it reaches tool 74.6%, command 35.0%, `is_write` 81.9%, `touches_poe_power` 80.8% on the 260 base rows with 71–84% of the mass on the labels for the choice questions and a 95% order-flip rate on the tool question (rotation averaging: 68.8→74.6%). As a gate: AUROC 0.879, 13 of 20 real confusions flagged, 36 of 151 correct calls false-flagged, 25 of 156 dangerous swaps accepted.

5.3 E3 — a fine-tuned 150M encoder

ModernBERT-base trained for 4 epochs on 2,605 examples (the training groups' rows and paraphrases; CE + Brier, augmentation 0.5) and evaluated packed on the 126 held-out rows: tool 70.6%, command 65.1% (65.9% with alternatives), `is_write` 81.7%, `touches_poe_power` 82.5%; 6 dangerous confusions, none above 0.9 confidence; `touches_poe_power` ECE 0.129→0.015. The readout on the same 126 rows: 84.9 / 69.8 (73.0) / 90.5 / 96.0. Two recipe findings: the span head is position-aware — asked alone, a question trained in the second slot drops from 55% to 7% — so batteries must be evaluated as packed, or trained with question-order shuffling; and the gate head, trained with a 1:2 positive:negative mix, learned the prior (it accepts 6.7% of gold proposals). A DeBERTa-v3-base run under the same recipe diverged to NaN confidences under bf16 autocast (a known DeBERTa-v3 failure mode) and is not reported.

Round 2. Shuffling the question order in training (each training example emitted once more with its four questions permuted) and training for 8 epochs changes the picture: on the same 126 held-out rows the encoder reaches tool 88.1%, command 77.8% (79.4% with alternatives), `is_write` 95.2%, `touches_poe_power` 96.8%, with ECE 0.047→0.021 (tool) and 0.069→0.058 (command); it is now the better router than the 4B readout on every question, with 3 dangerous confusions (1 above 0.9 after scaling), at 126 ms per NOUL on the board's CPU (Table 3). The gate does not follow: a dedicated model trained on 2,080 class-balanced pairs for 6 epochs reaches AUROC 0.556 on held-out pairs (0.68 on the real ones) and accepts 79% of wrong-port proposals, which locates the failure — the gate requires matching the proposed port and value against the request, and this encoder does not learn exact matching from two thousand synthetic pairs. The decoder readouts do.

5.4 E4 — latency

Table 3. The board runs the Snapdragon `llama-server` build CPU-only beside the production NPU tenant (the binary links the Hexagon backend and aborts on the busy DSP unless `GGML_HEXAGON_NDEV=0`); the encoders run under CPU torch in an isolated environment. With the question legend prefix-cached, a readout router question costs 1.23–1.32 s wall on six Kryo threads for 20 evaluated tokens (≈16 tok/s incremental prefill) and the gate question 3.1 s for 52; a four-question battery is therefore ∼5 s, sequential, while one gate check per write is ∼3 s. The fine-tuned 150M encoder answers one NOUL in 126 ms at four threads and the gate question in 183 ms; its packed four-question battery is a ∼1,100-token sequence and costs 2.5 s. Co-tenancy costs nothing measurable: a generative turn through the NPU runtime (96-token answer) took 4.43–4.52 s idle and 4.32–4.33 s while the CPU readout answered a continuous stream of questions.

backend	host	4-question battery	one NOUL	gate NOUL
readout Qwen3-4B, prefix cached	QCS9075 CPU, 6 thr.	≈5.1 s	1.24 s	3.1 s
readout Qwen3-4B, prefix cached	Grace CPU, 12 thr.	≈2.4 s	0.56 s	1.32 s
fine-tuned ModernBERT-base	QCS9075 CPU, 4 thr. (1 thr.)	2.51 s (6.47)	126 ms (195)	183 ms (332)
fine-tuned ModernBERT-base	GB10 GPU	12 ms	9 ms	—
Laya 421M	QCS9075 CPU, 4 thr. (1 thr.)	3.37 s (5.47)	352 ms (720)	527 ms (1199)
openJev-verdict 151M, torch (ONNX)	QCS9075 CPU, 4 thr.	3.35 s (4.75)	295 ms (361)	337 ms (429)
NLI DeBERTa-v3-base	QCS9075 CPU, 4 thr.	5.15 s	274 ms	330 ms
open-jev-deberta-v3-large	QCS9075 CPU, 4 thr.	5.79 s	799 ms	982 ms

Table 3: Median wall time per call. Encoder batteries answer all four questions in one packed pass; the 24-option question dominates their length.

6 Serving-path reality

The production runtime on the board (GenieX v0.4.0, compiled W4A16 bundle) answered a request with `logprobs:true, max_tokens:1` by returning `"logprobs": null`, three generated tokens, and `cached_tokens: 0`; the binaries of every GenieX version on hand contain no `n_probs` plumbing. On that path a typed decision collapses to a text label: measured live on the 195 paraphrases, a 4-way text-label tool choice from the same 4B scored 88.7% (75.4% on indirect phrasing) against 93.3% for its own tool call after format normalization, and a 24-way command label 70.8% against 83.1%; 7% of its answers changed when the option order was reversed. Probabilities are available only through upstream llama.cpp on the Hexagon backend — built, not yet deployed, and unable to share the NPU with the production tenant — or on the CPU, where the readout and every encoder above run beside the NPU agent. The CPU latency numbers in §5 are the operative ones.

7 Discussion

What the gate buys. A generated tool call gives the human a command and a reason. The gate question gives them a number whose meaning has been measured: at the 0.5 operating point, three of four dangerous misreads carry a warning and one in five correct proposals does. Whether that is the right point is a product decision — the risk–coverage curve is in the repository — and the substantive change is that it *is* a decision. Nothing here loosens the gate; the design rule that no device write executes on a probability is kept.

Readout vs encoder. On the router the fine-tuned encoder wins once its training shuffles the question order: better on every question than the 4B readout on identical held-out rows, calibrated as shipped, and forty times cheaper per question on the board's CPU. On the gate only the decoder readouts work; every encoder variant, including one trained on balanced gate pairs, stays near chance, and the wrong-port acceptance rate says why (exact port and value matching is not learned from two thousand synthetic pairs). The division of labour on this class of hardware is therefore concrete: an encoder for the per-turn router and pre-check (milliseconds), a decoder readout for the per-write gate (seconds, once per confirmation), and a llama.cpp server on the CPU beside the production runtime to serve the latter.

Calibration is measured, not assumed. Every backend here was over-confident before scaling, the readout's operating point moved by 0.6 between two phrasings of one question, and the encoder's answers depended on where the question sat in the pack. The transferable rule is procedural: label a few hundred of your own rows, group them by seed, fit the temperature out-of-fold, and publish the reliability diagram beside the accuracy.

Limitations. The authored rows and training paraphrases were LLM-generated and LLM-verified; human review is pending and could move the adversarial numbers. There is one domain and N is in the hundreds, so per-style differences of a few points are within noise. We could not run cloud Jev (no key); its published calibration is a vendor claim we neither confirm nor dispute. The fine-tuning recipe is one configuration of one library; round 2 fixes two identified defects rather than searching the space.

8 Related work

Jev and its documentation [11, 13, 12]; open reproductions [5, 8, 6, 1, 3] and the typed-decisions benchmark [7]; calibration of neural networks [4] and of instruction-tuned LMs; multiple-choice order sensitivity [16, 10]; zero-shot classification with NLI cross-encoders [15]; selective prediction [2]; tool-calling evaluation [9]; the edge deployment this work extends [14].

9 Conclusion

The interface Jev popularised — typed questions, a distribution per question, a confidence axis — can be served on an air-gapped board today, from the very model whose calls it checks, and it gives a human confirm gate something it did not have: a calibrated, measured probability that the proposed command is what the operator asked for. It does not make the small model a better router, it is not calibrated until it is calibrated on the deployment's own rows, and it costs seconds on a CPU. A 150M encoder fine-tuned on two thousand rows, with the question order shuffled, out-routes it at a fortieth of the cost but cannot learn the gate. The dataset, code and receipts are released so the next deployment can measure instead of assume.

References

[1] Zefan Cai. Open-jev: Lora adapters and scalar decision heads for typed decisions on qwen3.5. https://huggingface.co/ZefanCai/Open-Jev-2B; https://github.com/Zefan-Cai/Open-Jev, 2026.

[2] Yonatan Geifman and Ran El-Yaniv. Selective classification for deep neural networks. In *NeurIPS*, 2017.

[3] GitHub30. Openjev: open-weight system one model (jev-compatible) by candidate log-likelihood readout. https://github.com/GitHub30/OpenJev, 2026.

[4] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *ICML*, 2017.

[5] Heman10x. Openjev (verdict) 2.0: a calibrated 151m non-autoregressive decision engine. https://huggingface.co/heman10x/rlcd-modernbert-151m; https://github.com/Heman10x-NGU/Verdict-open-jev, 2026.

[6] Kotoba Labs. open-jev-deberta-v3-large: an open, jev-shaped typed-decision model. https://huggingface.co/com-kotobalabs/open-jev-deberta-v3-large; https://github.com/kotoba-lang/typed-decisions, 2026.

[7] LocalLLaMA. Localllama/typed-decisions: a benchmark of typed enterprise decisions. https://huggingface.co/datasets/LocalLLaMA/typed-decisions, 2026.

[8] Nandha Kishor M and Convai Innovations. Laya: multilingual non-autoregressive system 1 decision model trained with rlcd. https://huggingface.co/convaiinnovations/laya; https://github.com/NandhaKishorM/laya, 2026.

[9] Shishir G. Patil et al. Berkeley function calling leaderboard. https://gorilla.cs.berkeley.edu/leaderboard.html, 2024.

[10] Pouya Pezeshkpour and Estevam Hruschka. Large language models sensitivity to the order of options in multiple-choice questions. *arXiv:2308.11483*, 2023.

[11] TypeSafe AI. Introducing system one models & jev. https://typesafe.ai/blog/introducing-system-one-models-and-jev, 2026. Released 2026-09-15.

[12] TypeSafe AI. Jev 1.13 jagged edges. https://docs.typesafe.ai/model-jaggedness/jev-1.13, 2026.

[13] TypeSafe AI. Typesafe documentation: primitives, confidence, patterns, models. https://docs.typesafe.ai/, 2026.

[14] Krish Viswanathan. What survives the edge: tool-calling llm agents on embedded npus (engineering report and paper outline), 2026. Measurements on a Qualcomm QCS9075 EVK; blog series at https://kvapr.github.io.

[15] Wenpeng Yin, Jamaal Hay, and Dan Roth. Benchmarking zero-shot text classification: Datasets, evaluation and entailment approach. In *EMNLP*, 2019.

[16] Chujie Zheng, Hao Zhou, Fandong Meng, Jie Zhou, and Minlie Huang. Large language models are not robust multiple choice selectors. In *ICLR*, 2024.